

Econ 630 part 2, Problem Set 5

Question 4

Let player 1 and 2's strategies be $(\sigma_U \circ U, (1 - \sigma_U) \circ D)$, and $(\sigma_L \circ L, (1 - \sigma_L) \circ R)$ respectively.

(a) For game 1, $E\pi_1(U|G1, \sigma_L) = \sigma_L$, $E\pi_1(D|G1, \sigma_L) = 0$. Therefore,

$$\sigma_U(\sigma_L) \begin{cases} = 1, & \text{if } \sigma_L > 0; \\ \in [0, 1], & \text{if } \sigma_L = 0. \end{cases}$$

Similarly, $E\pi_2(L|\sigma_U) = 0.5\sigma_U$, $E\pi_2(R|\sigma_U) = 1 - \sigma_U$. Therefore,

$$\sigma_L(\sigma_U) \begin{cases} = 1, & \text{if } \sigma_U > \frac{2}{3}; \\ \in [0, 1], & \text{if } \sigma_U = \frac{2}{3}; \\ = 0, & \text{if } \sigma_U < \frac{2}{3}. \end{cases}$$

We can draw the best response function as follows:

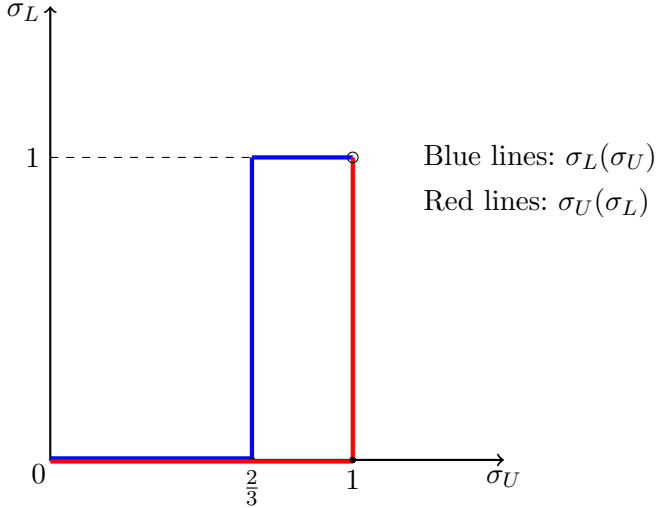


Figure 1: Game 1

The intersections of best response curves are the BNE:

$$(\sigma_U, \sigma_L) = (1, 1), \text{ or } \{\sigma_U \in [0, \frac{2}{3}], \sigma_L = 0\}.$$

For game 2, $E\pi_1(U|G2, \sigma_L) = 0$, $E\pi_1(D|G2, \sigma_L) = 2 - 2\sigma_L$. Therefore,

$$\sigma_U(\sigma_L) \begin{cases} = 0, & \text{if } \sigma_L < 1; \\ \in [0, 1], & \text{if } \sigma_L = 1. \end{cases}$$

Player 2's best response function is the same as game 1 in (a). We can draw the best response function as follows:

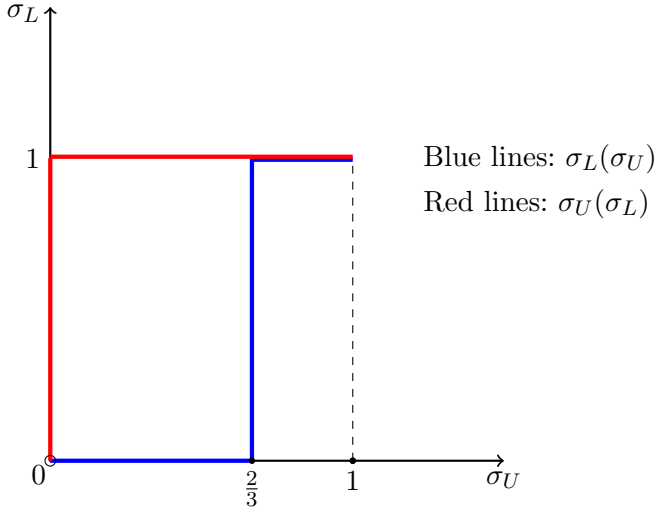


Figure 2: Game 2

The intersections of best response curves are the BNE:

$$(\sigma_U, \sigma_L) = (0, 0), \text{ or } \{\sigma_U \in [\frac{2}{3}, 1], \sigma_L = 1\}.$$